MIL-E-1/275C
10 March 1976
SUPERSEDING
MIL-E-1/275B
31 May 1968

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY

TYPES 5LP1A, 5LP5A, 5LP7A, AND 5LP11A

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Electrostatic deflection and focus

DIMENSIONS AND PIN CONNECTIONS: See figure 1

ABSOLUTE RATINGS:

Parameter:	Ef	Ec1	ed	Eb1	Eb2	Eb3	Rg	Zd	Eb3/Eb2	Alt
Unit:	V	Vdc	v	Vdc	Vdc	Vdc	Meg	Meg	Ratio	ft
Maximum:	6.9	0	550	1,100	2,200	4,400	1.5	1.0	2	10,000
Minimum:	5.7	-125	---	---	1,100	1,100	---	---	---	---
TEST CONDITIONS:										
P1A, P5A, P11A:	6.3	Adj	---	Focus	1,500	3,000	---	---	---	---
P7A:	6.3	Adj	---	Focus	2,000	4,000	---	---	---	---

GENERAL:

Qualification - Required

© denotes changes

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METHOD	REQUIREMENT OR TEST	TYPE	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Quality conformance inspection, part 1							
5106	Bulb, screen, and faceplate quality	All	-		---	---	---	---
5201	Voltage breakdown	All	-		---	---	---	---
5201	Voltage breakdown (electrostatic types)	All	-		---	---	---	---
5221	Light output	P1A	-		Light	4.0	---	ftL
		P5A	3		Light	0.65	---	ftL
		P11A	3		Light	2.0	---	ftL
5231	Spot position (electrostatic deflection)	All	-		---	---	20	mm
5231	Spot displacement (leakage)	All	-		Displ	---	5	mm
5223	Modulation	P1A	-	Light = 4.0 ftL	ΔEc1	---	35	Vdc
		P5A	3	Light = 0.65 ftL	ΔEc1	---	35	Vdc
		P7A	-	Ib3 = 10 μAdc	ΔEc1	---	45	Vdc
		P11A	3	Light = 2.0 ftL	ΔEc1	---	35	Vdc
5241	Grid cutoff voltage	P1A, P5A, P11A	-		Ec1	-22.5	-67.5	Vdc
5251	Grid No. 1 leakage current	All	-		---	---	---	---
5251	Anode No. 2 leakage current	All	-		---	---	---	---
5206	Gas "cross"	P1A	-	Light = 4.0 ftL	---	---	---	---
		P5A	3	Light = 0.65 ftL	---	---	---	---
		P7A	-	Ib3 = 10 μAdc	---	---	---	---
		P11A	3	Light = 2.0 ftL	---	---	---	---
	Quality conformance inspection, part 2							
5101	Side terminal and base alinement	All	-		---	---	---	---
5101	Base alinement (electrostatic types)	All	-	4D3; pin No. 6	---	---	---	---
5101	Neck and base alinement (electrostatic types)	All	-		---	---	---	---
5101	Side terminal alinement (electrostatic types)	All	-	3D4	---	---	---	---
1301	Heater current	All	-		If	540	660	mA
5101	Angle between traces	All	-		---	---	---	---
5251	Heater-cathode leakage current	All	-		---	---	---	---
5201	Electrode current (anode No. 1)	All	-	Ec1 = 0	Ib1	-50	10	μAdc
5201	Electrode current (cathode)	P1A	-	Light = 4.0 ftL	Ik	---	1,000	μAdc
		P5A	3	Light = 0.65 ftL	Ik	---	1,000	μAdc
		P7A	-	Ib3 = 10 μAdc	Ik	---	1,000	μAdc
		P11A	3	Light = 2.0 ftL	Ik	---	1,000	μAdc
5216	Stray light emission (conventional types)	All	-	Eb2 = 2,200 Vdc; Eb3 = 4,400 Vdc	---	---	---	---

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METHOD	REQUIREMENT OR TEST	TYPE	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Quality conformance inspection, part 2 -Continued							
5226	Line width "A" (electrostatic deflection)	P1A	1	Light = 4.0 ftL	Width	---	0.6	mm
		P5A	3	Light = 0.65 ftL	Width	---	0.6	mm
		P7A	-	Ib3 = 10 µAdc	Width	---	0.6	mm
		P11A	3	Light = 2.0 ftL	Width	---	0.6	mm
5226	Line width "B" (electrostatic deflection)	P1A	2	Light = 4.0 ftL	Width	---	0.7	mm
		P5A	3	Light = 0.65 ftL	Width	---	0.7	mm
		P7A	-	Ib3 = 10 µAdc	Width	---	0.7	mm
		P11A	3	Light = 2.0 ftL	Width	---	0.7	mm
5248	Deflection factor (1D2)	P1A, P5A, P11A	-		DF	62	93	Vdc/in.
		P7A	-		DF	82	124	Vdc/in.
5248	Deflection factor (3D4)	P1A, P5A, P11A	-		DF	54	81	Vdc/in.
		P7A	-		DF	73	109	Vdc/in.
5246	Focusing voltage at modulation condition	P1A, P5A, P11A	-		Eb1	282	475	Vdc
		P7A	-		Eb1	375	632	Vdc
5246	Focusing voltage at cutoff	P1A, P5A, P11A	-		Eb1	300	450	Vdc
		P7A	-		Eb1	400	600	Vdc
5221	Screens	P7A	-		---	---	---	---
1101	Secureness of base, cap, or insert	A11	-		---	---	---	---
(C) 1111	Base pin solder depth	A11	-		---	---	---	---
1105	Permanence of marking	A11	-		---	---	---	---
	Quality conformance inspection, part 3							
---	Life test	A11	-	Group C; Eb2 = 2,200 Vdc; Eb3 = 4,400 Vdc; t = 500 hours (min)	---	---	---	---
		P1A	-	Light = 4.0 ftL	---	---	---	---
		P5A	3	Light = 0.65 ftL	---	---	---	---
		P7A	-	Ib3 = 10 µAdc	---	---	---	---
		P11A	3	Light = 2.0 ftL	---	---	---	---
---	Life-test end points:							
5251	Heater-cathode leakage current	A11	-		---	---	---	---
5251	Grid No. 1 leakage current	A11	-		---	---	---	---
5251	Anode No. 2 leakage current	A11	-		---	---	---	---
5216	Stray light emission (conven-types)	A11	-		---	---	---	---
5226	Line width "A"	P1A	-	Light = 3.0 ftL	Width	---	0.6	mm
5226	Line width "B"	P1A	-	Light = 3.0 ftL	Width	---	0.7	mm
5223	Modulation	P1A	-	Light = 3.0 ftL	ΔEc1	---	35	Vdc
5226	Line width "A"	P5A	3	Light = 0.5 ftL	Width	---	0.6	mm
5226	Line width "B"	P5A	3	Light = 0.5 ftL	Width	---	0.7	mm
5223	Modulation	P5A	3	Light = 0.5 ftL	ΔEc1	---	35	Vdc
5226	Line width "A"	P7A	-	Ib3 = 7.5 uAdc	Width	---	0.6	mm
5226	Line width "B"	P7A	-	Ib3 = 7.5 µAdc	Width	---	0.7	mm
5223	Modulation	P7A	-	Ib3 = 7.5 uAdc	ΔEc1	---	45	Vdc
5226	Line width "A"	P11A	3	Light = 1.5 ftL	Width	---	0.6	mm
5226	Line width "B"	P11A	3	Light = 1.5 ftL	Width	---	0.7	mm
5223	Modulation	P11A	3	Light = 1.5 ftL	ΔEc1	---	35	Vdc

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METHOD	REQUIREMENT OR TEST	TYPE	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Quality conformance inspection, part 3 -Continued							
1216	Base material insulating quality	All	4	Zone 5 (min)	---	---	---	---
5216	Cathode illumination	All	4		---	---	---	---
1141	Pressure (implosion)	All	4		---	---	---	---
5248	Deflection-factor uniformity	All	4		---	---	---	---
1331	Direct-interelectrode capacitance	All	4	g1 to all	Cg1	---	13.0	pF
				D1 to D2	C1D2	---	3.0	pF
				D3 to D4	C3D4	---	2.5	pF
				D1 to all	CD1	---	14.0	pF
				D2 to all	CD2	---	14.0	pF
				D3 to all	CD3	---	9.0	pF
				D4 to all	CD4	---	9.0	pF
5101	Neck and bulb alinement (electro-static types)	All	4		Dia	---	1.87	inch
5111	Vibration	All	4		Width	---	1.0	mm
5248	Deflection factor (1D2)	P1A, P5A, P11A	4	Eb3 = Eb2 = 1,500 Vdc	DF	50	76	Vdc/in.
		P7A	4	Eb3 = Eb2 = 2,000 Vdc	DF	67	101	Vdc/in.
5248	Deflection factor (3D4)	P1A, P5A, P11A	4	Eb3 = Eb2 = 1,500 Vdc	DF	46	68	Vdc/in.
		P7A	4	Eb3 = Eb2 = 2,000 Vdc	DF	61	91	Vdc/in.

NOTES:

1. The deflecting plates shall be returned to No. 2 anode through a minimum of 2.5 megohm resistors. The light output shall be set at 2 ftL. The high-frequency scanning shall be applied to the deflecting plates nearest the screen and the amplitude shall be adjusted to give a line length of approximately 90 percent of the maximum tube diameter. The low-frequency scanning amplitude shall be expanded to approximately 90 percent of the maximum tube diameter in the direction perpendicular to the direction of high-frequency scanning. Readjustment may be made for best overall focus. The tubes shall be observed for deflection defocusing, astigmatism or spot ellipticity observable to the eye as evidenced by fuzziness due to lack of sharpness of trace (usually around edges), bow-tieing (irregular widths of any single line when observed at different points), bowing of trace other than that normally caused by curvature of bulb. This test for focus is to be made in addition to the line width measurements.

2. The same conditions shall be set up as described in note 1 except that the connection of deflecting elements to the low- and high-frequency scanning supplies shall be interchanged and the amplitudes adjusted to 90 percent of the maximum tube diameter in both directions without any adjustment of focus from note 1 conditions. An examination for defocusing, astigmatism, or spot ellipticity shall be made as in note 1.

3. As measured with a 2x2 inch raster using a type 3 Photronic cell without eye correction, calibrated in foot candles of illumination from a light source having a color temperature of 2,700° K.

(C) 4. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, coce level I, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 80
DSA - ES

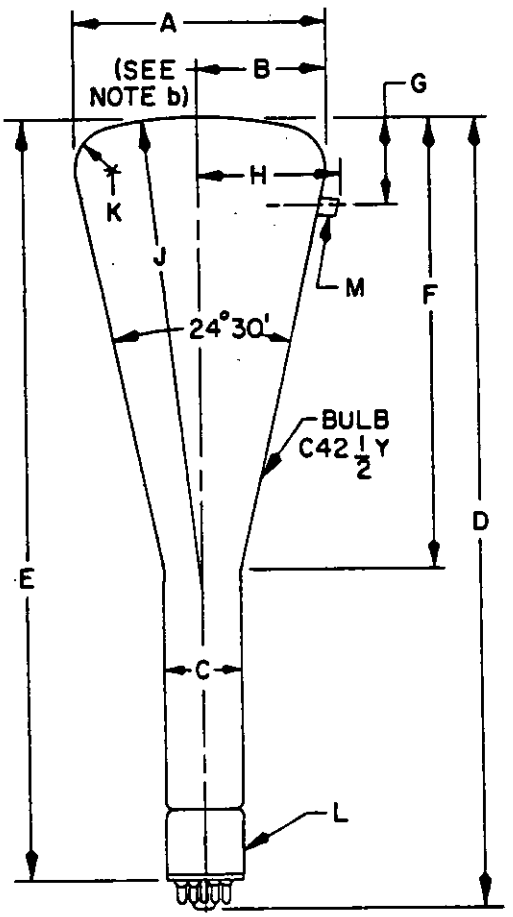
User activities:
Army-- MU
Navy - AS, OS, MC, CG
Air Force - 11

Preparing activity:
Navy - EC

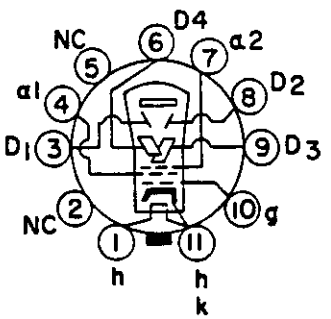
Agent:
DSA - ES

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses (See note a)	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	5.250 (133.35)	5.375 (136.53)
B	2.250 (57.15)	
C	1.562 (39.67)	1.688 (42.88)
D	16.375 (415.93)	17.125 (434.98)
F		9.500 (241.30)
G	1.500 (38.10)	2.000 (50.80)
H		3.375 (85.73)
Quality conformance inspection, part 3 (See note 4)		
L	Base: B11-66	
M	Bulb contact: C1-1	
Reference dimensions (See note c)		
E	16.220 (411.99)	
J	10.000 (254.00)	
K	.750 (19.05) R	



NOTES:
a. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
b. Useful screen radius.
c. Reference dimensions are provided for information only and are not required for inspection purposes.

FIGURE 1. Outline drawing of electron tube types 5LP1A, 5LP5A, 5LP7A and 5LP11A.

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